## CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The sum of the first 35 odd numbers is 1225. The sum of the odd numbers from 31 to 69 is: 1

**(a)** 1210 **(b)** 1000 **(c)** 400 **(d)** 969

**2.** The difference between the 25th and the 24th square numbers is: 1

**(a)** 25 **(b)** 51 **(c)** 49 **(d)** 47

**3.** A pattern adds the same number every time. Its first term is 12 and its 17th term is 220. The number added each time is: 1

**(a)** 12 **(b)** 15 **(c)** 13 **(d)** 14

**4.** A pattern adds 1, then 2, then 3, ... Starting at 25, the term after 15 additions is: 1

**(a)** 145 **(b)** 130 **(c)** 250 **(d)** 40

**5.** The sum $1^3 + 2^3 + \cdots + 10^3$ equals: 1

**(a)** 3025 **(b)** 1000 **(c)** 110 **(d)** 55

**6.** The 57th triangular number minus the 56th triangular number is: 1

**(a)** 57 **(b)** 56 **(c)** 58 **(d)** 826

**7.** At a party, 20 people each hug every other person exactly once. Number of hugs: 1

**(a)** 200 **(b)** 380 **(c)** 190 **(d)** 210

**8.** If $1 + 2 + 3 + \cdots + n = 1378$, then $n$ is: 1

**(a)** 51 **(b)** 53 **(c)** 26 **(d)** 52

**9.** **Assertion (A):** The sum $1 + 2 + 3 + \cdots + 10$ is 55. 1

**Reason (R):** The sum of the first $n$ natural numbers is $n \times n$.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** The sequence $2, 4, 8, 16, \ldots$ has a common difference. 1

**Reason (R):** A sequence in which the same number is added every time has a common difference.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** The sum of the first $n$ odd numbers is 1444. Find $n$ and the 38th odd number. 2

**12.** Find the sum of odd numbers from 23 to 55. 2

**13.** 17 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

**14.** The sum of the first $n$ odd numbers is 1849. Find $n$ and the 43rd odd number. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** In a pattern the 1st term is 6, the 2nd is 19, and so on by adding 13. Which term of the pattern equals 409? Find also the term just before it. 3

**16.** Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 59. Then find the sum of the odd numbers from 61 to 75. 3

**17.** A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 8, and how many in the first 4 rows together? 3

**OR**

In a pattern the 1st term is 31, the 2nd is 43, and so on by adding 12. Which term of the pattern equals 355? Find also the term just before it. 3

**18.** Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 69. Then find the sum of the odd numbers from 71 to 83. 3

**OR**

A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 9, and how many in the first 5 rows together? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A fruit seller stacks oranges in a triangle. The top row has 1 orange, the second row has 2, the third row has 3, and so on. 5

(i) How many oranges are in a stack of 6 rows?

(ii) How many oranges are in a stack of 8 rows?

(iii) How many oranges are in the 10th row alone?

(iv) The seller has 50 oranges. What is the biggest full triangle he can make, and how many oranges are left over?

**20.** In a pattern of dot figures, figure 1 has 3 dots and each next figure has 11 more dots than the one before. How many dots are in figure 15? Which figure has 212 dots? 5

**OR**

Two sequences: A is $11, 17, 23, 29, \ldots$ and B is $6, 14, 22, 30, \ldots$. Find the 14th term of each and decide which is larger and by how much. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**